Q. KNN (K Nearest Neighbors) Classification:-

Remember

1. What is KNN?

=> • KNN = K-Nearest Neighbors

• It is a supervised learning algorithm.

• used mostly for classification (can also be used for regression). (non-linear classification)

• Very simple & intuitive method: "Look at your closest neighbors"

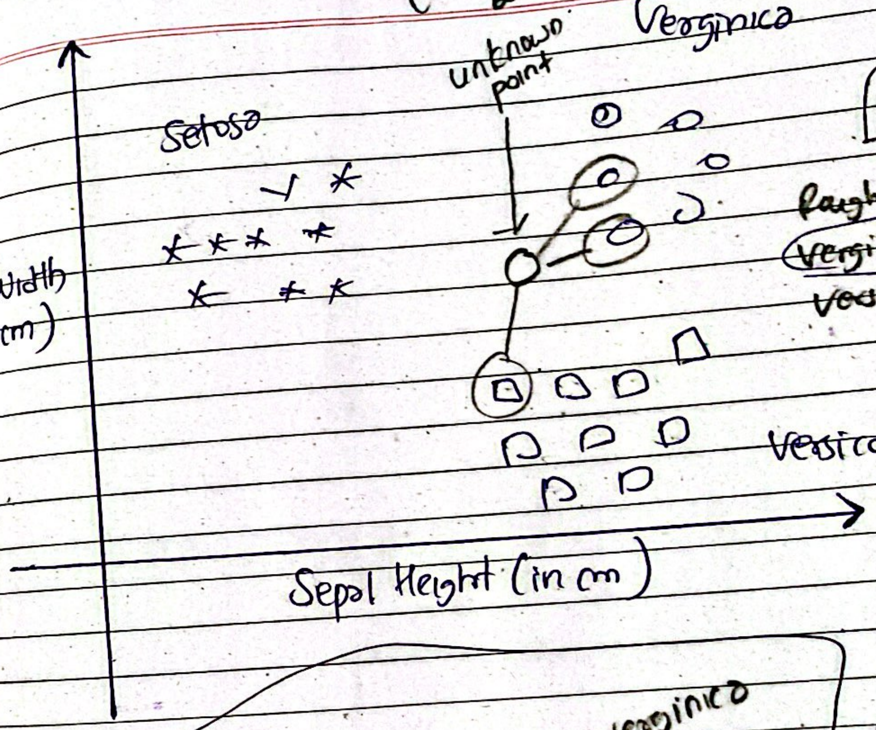
2. How KNN works (concept)

1. You have a dataset with labels (e.g. types of iris flowers).
2. When a new point comes, you: (most near 3 points)
   - choose a value of K (example: K=3 or 5)
   - find the K nearest points in the training data
   - Use Euclidean distance to measure closeness.
3. Majority vote among those K neighbors determines the class.

Ex:-
- New point is yellow
- K=3 → check 3 nearest ~~neighbors~~
- If neighbors are: Virginica, Virginica, Virginica → Prediction = Virginica

Analysis (which class will it belong to)

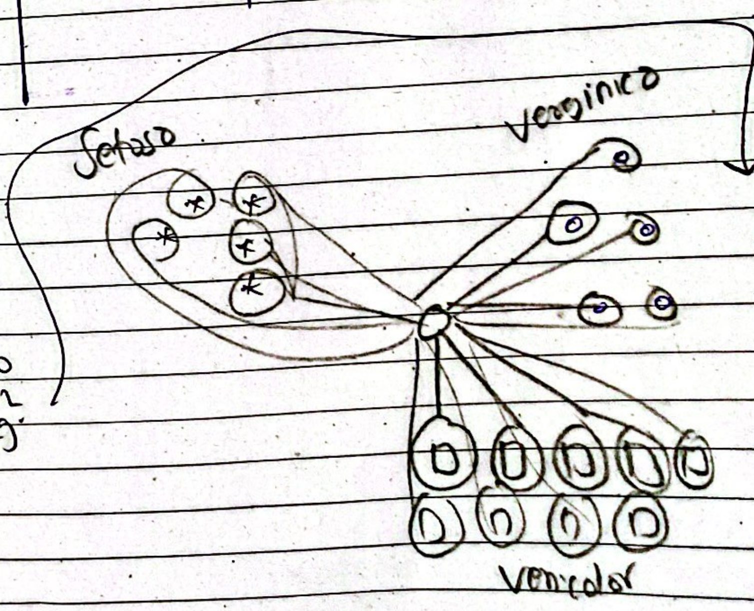
Choosing K



K=3

Rough calc:
Verginica → 2
Versicolor → 1

Where things could go wrong.



Rough calc:

K=20

Setosa → 5
Verginica → 6
Versicolor → 9

So, choose K wisely

Also, check for imbalanced classes data.

3. Choosing k
- No fixed rule for choosing k
- Usually small numbers like 3 or 5.
- If:
  - k too small → noisy, overfitting
  - k too large → may mix other classes, misclassification.

Example:-
- k=20 may give wrong result if class distribution is unbalanced.

4. KNN with more features
- In example, only 2 features: sepal width & sepal height.
- KNN works fine even with n features (multi-dimensional space).
- We can't visualize them, but math still works.

5. Iris Dataset
- comes from sklearn.datasets
- has 4 features
  1. sepal length
  2. sepal width
  3. petal length
  4. petal width

- Classes:
  - 0 – Setosa
  - 1 – Versicolor
  - 2 – Virginica

Total samples = 150 (50 per class)

6. Train/Test Split
- Usual practice: – split into training & testing
- Example:-
  - Training samples = 120
  - Test Samples = 30

Imp

7 – Building a KNN model (Python Steps)

1- load dataset

```
from sklearn.datasets import load_iris
iris = load_iris()
```

2- Make Dataframe
- add features + Target Column

3. Split into train/test

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(...)
```

Analysis

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4. Create KNN classifier

```
from sklearn.neighbors import KNeighborsClassifier
model = KNeighborsClassifier(n_neighbors=3)
```
(K)

5. fit the model

```
model.fit(X_train, y_train)
```

6. Calculate accuracy

```
model.score(X_test, y_test)
```

8. Euclidean Distance (Default Metric)
- In sklearn, `metric = "minkowski", p=2`

=) This means Euclidean distance

9. Confusion Matrix
- tells which classes were predicted correctly or incorrectly.

Steps:
1. predict on test data:

```
y_pred = model.predict(X_test)
```

2. Create confusion matrix:

```
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
```

3- Heatmap (if using seaborn):
- Diagonal values = correct predictions.
- Off-diagonal = wrong predictions

~~Example meaning:~~ -

- ~~Value at~~

10- Classification Report

- Shows:
  - Precision
  - Recall
  - f1-score
  - for each class

Imp (after confusion matrix)

```
from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred))
```

11. Important Concepts
- Accuracy
- Precision & Recall.

Date
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Choosing best value of K

You can find the best k using:-

- Trial & error (manual)
- GridSearchCV
- K-fold Cross Validation

Short Summary (Easy to Revise)

- KNN = simple, intuitive classification method.
- find k nearest neighbors → take majority class
- choose k carefully.
- Uses Euclidean distance by default.
- Easy to implement in sklearn.
- Evaluate using accuracy, confusion matrix, classification report.